Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Spring 2025
Laboratory Exercise 5
Layout Design Library

Part 1 Due: 02/21/2025
Part 2 Due: 02/28/2025

Contents

1 Introduction

The goal of this lab exercise is to design a logically complete standard cell library. This involves creating layouts for the already existing cells: 2 input NAND, 2 input NOR, 2-1 MUX, 2 input XOR, and 3 input AOI as well as getting post-layout test results. For this lab, you will complete the datasheets that you submitted for Lab3 by adding in the layout captures for each gate.

There is no lab report for this lab. Complete the worksheet from section 6, and update the 5 datasheets with the layout captures. Submit the worksheet, updated datasheet, and the signoff sheet to lab dropbox.

Keep the following in mind as you work through this lab.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES ARE LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

  /classes/ee620/bin/unlock_cadence.csh

- **Hotkeys:** You should learn how to use the hotkeys for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

  Options → Bindkey

  Select a category (e.g. **layout**) and a list of current hotkeys (bindkeys) will be shown for it.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!
- DO NOT delete any Cadence files from command line or file manager!

2 Prelab - Stick Diagrams

Before coming to the lab, draw the stick diagram for the gates you designed in Lab3 and Lab1 (i.e. NAND. NOR, XOR, MUX, and AOI).

All stick diagrams must be colored, properly labeled, and made using computer software (**not hand-drawn**).

3 Procedure

Draw the stick diagrams for all your schematics; NAND, NOR, MUX, XOR and AOI. Now that you have the blue print of how your layout designs should look, you can proceed to creating the layouts.

When starting this lab exercise, make sure you are on a fresh logged in session. If not, then close Cadence, log out and log back in. Open the Cadence tools and from the Library Manager window, select your library, then a cell (start with NANDX2). Then select **File** → **New** → **From Cellview**.

Then, use your stick diagram to design your layout.

Your layout must follow all of the provided guidelines! We can not provide signoffs on a layout that fails to meet them. There is a lot to digest, make sure you fully understand the guidelines below before moving on.

These are the standard cell layout guidelines to follow:

- $\lambda = 25nm$ or $0.025\mu m$
- All cells will be starting with the parametric layout template you learned in Lab Exercise 4.
  - PR boundary Height is 1.71μm
  - PR boundary Width is a multiple of $8\lambda = 0.2\mu m$ (the number of gates will determine the amount of poly-silicon you need which will make your design wider)
  - The lower left hand corner PR boundary of all cells should be placed at the **Origin** (X-axis, Y-axis = 0,0)
- I/O and VDD/VSS Pins:
  - I/O Pins (and labels) on **Metal 2— Pin only** *You will need to draw a* ***vertical*** *metal 2 strip surrounding the metal2 pin for future auto routing. Ensure that the area of metal2 is greater than 0.2 μm^2 otherwise DRC violations will occur.*
    - An example of I/O pin made with Metal2 is shown in Figure 1

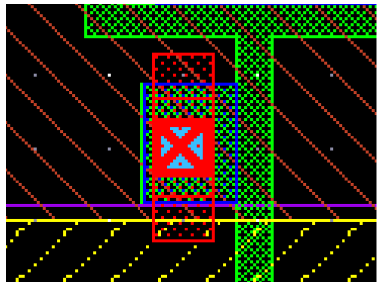

Figure 1: I/O M2 Pin Example

  - VDD/VSS Pins (and labels) must be made with **Metal 1— Pin only.**
  - VDD pins must be placed at the top of the cell (i.e. the power rail), and VSS pin must be placed at the bottom of the cell (i.e. the ground strip)
  - For labeling purposes, recall the special treatment of VDD/VSS as these are global signals VDD! and VSS!
  - Each row and each column can only have **one** I/O pin

- **The location of your I/O pins matter**. Each square measures 4λ. Notice that there are bolder grid-lines and lighter ones. The bolder grid-lines encapsulate 4 of the 4λ squares, making them an 8λ x 8λ grid. Your pins should fall within the PR boundary and in the center of horizontal (x-direction) and vertical (y-direction) minor axes (darker grey grid lines) or channels with a spacing of 8λ. In other words, I/O pins must be placed on the grid. An example of valid I/O pin locations is shown in Figure 2.

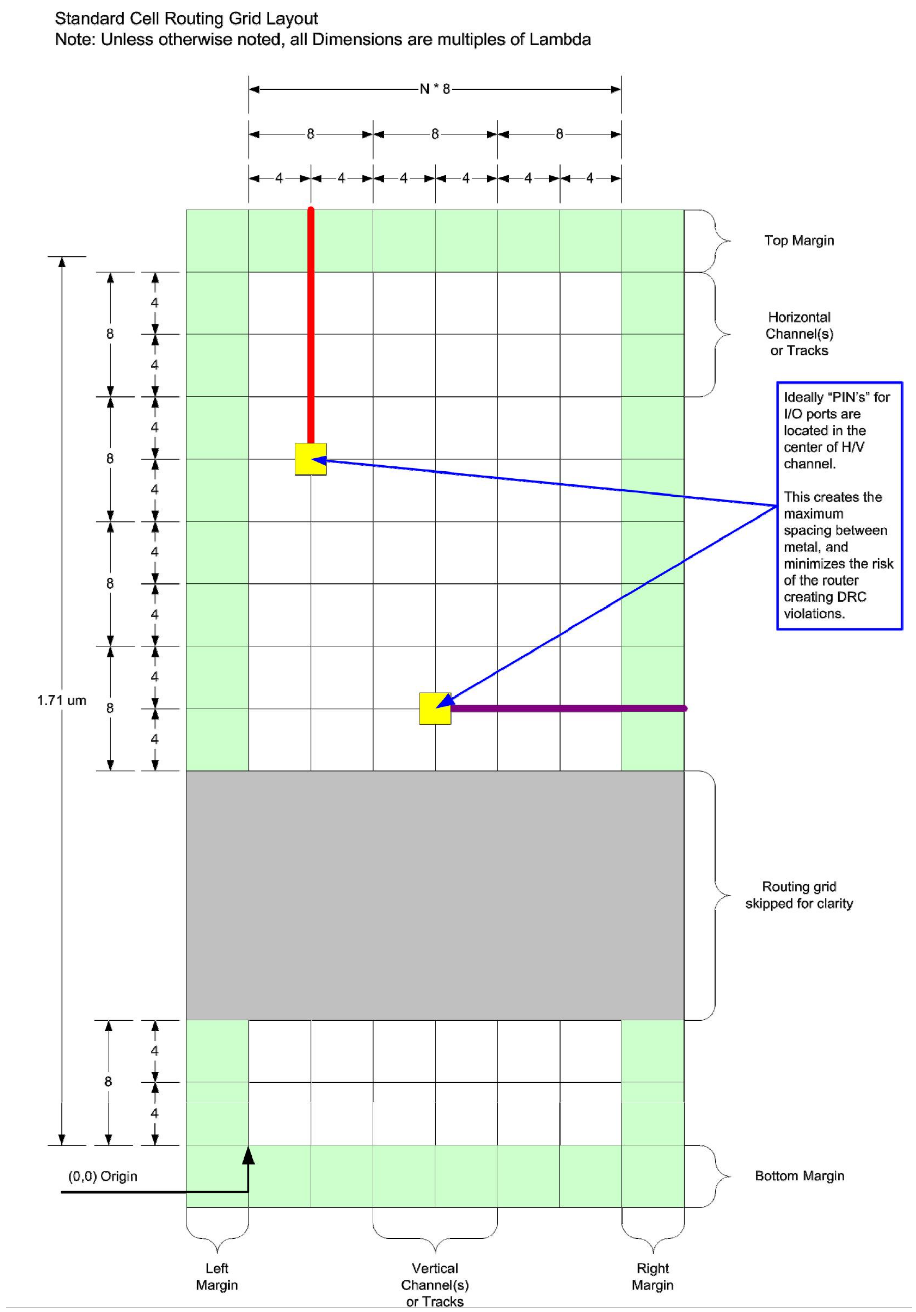

Figure 2: Pin Placement Routing Grid Layout

- A cell is designed using an 8λ grid and has the first possible I/O port location in the center of the first full vertical routing track - 4λ inside the left edge of the PR Boundary.
- **The final width of the cell PR Boundary MUST be a multiple of** 8λ
- Cell designs **will be restricted to using the following layers:**

  - Diffusion (any direction, minimal length)
  - Poly (any direction)
  - M1 (any direction)
  - M2 (vertical only) only for I/O pins
  - Note that the auto-router used in later labs will be routed using M2(vertical direction) and higher metal layers (e.g. M3, M4...up to M11). **We want to prevent using M2 in routing so that it gives the auto-router more room to route and connect pins.**

- When using contacts, we must use ≥ 2 for any connection. More contacts reduce resistance, therefore, more contacts leads to more efficient devices. You should strive to use as many contacts as you can reasonably fit in your devices.

- Remember that the length of the poly that overlaps with the oxide is the width of the transistor

- If the diffusion layer is not long enough to fit 2 contacts, you can expand the diffusion to allow more than 1 contact. These are called **hats**

  - Diffusion/active has to be top/bottom symmetrical. For example, consider the **hats** and symmetry shown in Figure 3

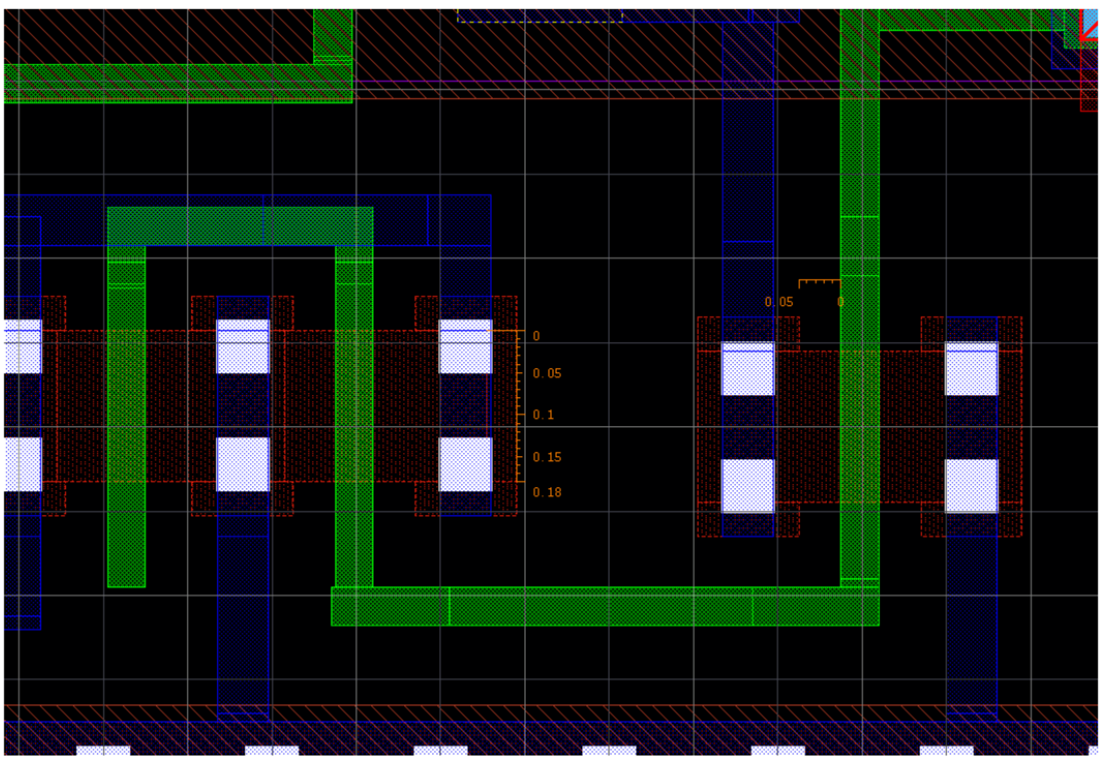

Figure 3: Diffusion Region Extensions (Hats) for Contacts

- When using M2 you have to use vias instead of contacts. Vias are essentially a bridge (or tunnel) that connects layers. In this case, M2 to M1. To create a via, go to **Create → Via** or press **O** on the keyboard and make sure the form matches with the following image. **Note that vias can also be used between other layers such as poly and oxide.**

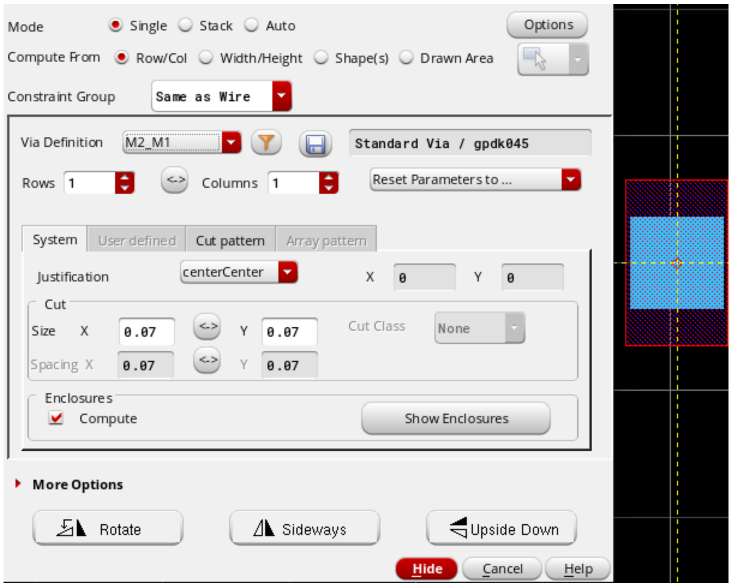

Figure 4: Crate Vias Form

- Make sure your design passes the abutment test. This means that two standard cells can be placed adjacent to each other without issues
  - Make sure all Oxide, Metal1, and Poly layers are placed at least 0.04 um from the PR boundary
  - This is technically not a design rule violation that is reported during DRC, however, it will cause you A LOT OF trouble when using the auto router

You should be using the GPDK design rules PDF posted in MyCourses to guide you through this process.

At this point, make sure that you understand the GPDK library design rules and practices. Ask the TAs or lab instructors for clarification if there is any confusion.

Please read the next page entirely before you get started with the layout.

3.1 DRC Tips

You should periodically run DRC as you make progress on your layout instead running it only at the very end. Another handy tool is to have Active DRC turned on , this will show you DRC errors in real-time as you place each material. Use this to your advantage as well as the GPDK Design Rules manual to minimize errors during the layout design process.

3.2 LVS Tips

When running LVS, you will encounter brand new errors that are more challenging to interpret than those you have previously seen. To better understand these errors you can use the Graphical LVS Debugger tool. This tool will show you exactly where your layout diverged from the schematic. This feature can be found on the LVS Debug Environment window at the top **Tools** → **Graphical LVS Debugger**.

After you select the Graphical LVS Debugger, a new window will pop out showing all the routing and comparing the schematic to the layout design. You will see a legend at the bottom left corner that looks like this:

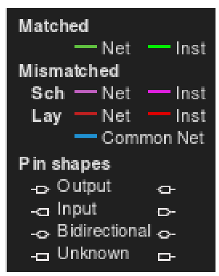

Figure 5: Graphical LVS Debugger

The legend will guide you on what each wire means. Head to the schematic comparison view and click on a wire (make sure it is highlighted after you click) then press 'P'. A window will pop up with either the layout or schematic window, depending on which wire you selected. The error will be highlighted in magenta.

Use minimum distances/dimensions as much as possible! The height is fixed, therefore, you must carefully consider how to fit all the necessary layers.

4 RC RCX & Testing

Use Lab 4 as a reference on how to perform testing using the "av_extraction" file to test your layouts. You will repeat the same tests you made for Lab 3, but with extracted parasitic. You should not have to make brand new tests. Instead, re-use the tests you have already created.

5 Closing Remarks

Congrats! You have created 5 more layouts! Now, complete the worksheet (see next page) by answering all of the questions. The deliverables of this lab are:

- Signoff sheet with all of the items completed
- Updated datasheets for NAND. NOR, MUX, XOR, and AOI (from Lab3). Update them by filling out the layout width, height, and area. Don't forget to include a capture of your layout in the datasheets
- Worksheet with all of the questions answered

6 Worksheet

Name:

1. What are some techniques (or something that you can add/reduce) to reduce the parasitic capacitance? *Note: This could include several different methods. There are many correct answers to this.*

Hint: Think about the RC delay model...

2. Remember our height constraint in the layout - $1.71\mu m$. Consider a scenario where we are now working with a design that does not fit in the cell template.

a. What technique should we apply to the design to allow us to fit it in the cell template? (Hint: folding...)

b. Draw a stick diagram of *any design of your choice* **before and after** applying the above technique.

7 Signoff Sheet

You do not need to have the 'Worksheet' and 'Datasheet' sign-off before the lab submission deadline, it will be graded after it is submitted.

Table 1: Layout Design Library

Students Name: Section:

Prelab	Point Value	Points Earned	Date
All 5 Stick Diagrams	10		
Part 1	Point Value	Points Earned	Date
NAND: DRC, LVS, RCX	12		
NOR: DRC, LVS, RCX	12		
AOI: DRC, LVS, RCX	12		
NAND, NOR, AOI Tests	8		
Part 2	Point Value	Points Earned	Date
MUX: DRC, LVS, RCX	12		
XOR: DRC, LVS, RCX	12		
MUX & XOR Tests	6		
Worksheet	8		
Datasheets	8		
Total Points	100		